Loss Aversion Shifts the Coordination Boundary in Cooperative MARL: A Phase Diagram Analysis in a Multi-Echelon Supply Chain

Agriya Yadav*

June 13, 2026

Abstract

Cooperative multi-agent reinforcement learning (MARL) systems fail when reward centralization falls below a critical threshold: agents converge to a zero-order *lazy-agent* policy, abandoning cooperation entirely. Despite widespread recognition of this failure mode, the collapse boundary has not been characterized as a quantitative, measurable object. The threshold is not a fixed constant—it shifts with training horizon (metastability) and, as we show, with agent-level prospect-theoretic loss aversion (λ), making it insufficient to study reward structure or behavioral bias in isolation. We construct the first quantitative phase diagram of cooperative MARL coordination collapse in (reward centralization w, training horizon T, loss aversion λ) space, using pre-registered collapse definitions and bootstrap-estimated critical points with confidence intervals. Across 1,124 controlled experiments on a multi-echelon supply chain substrate, we identify $w_{\text{crit}} = 0.666$ $[0.635, 0.694]$ at 300k steps under no behavioral bias. A three-way null-model decomposition reveals that probability weighting alone (α=β=0.88, λ=1) shifts w_{crit} *left* by 0.100 $[0.033, 0.172]$—facilitating coordination—while loss aversion at λ=3 and λ=7 shifts it *right* by 0.089 $[0.041, 0.141]$ and 0.166 $[0.135, 0.202]$ respectively. Probability weighting and loss aversion exert statistically significant effects in *opposite directions*, a decomposition that, to our knowledge, has not previously been demonstrated in cooperative MARL.

1 Introduction

Cooperative multi-agent reinforcement learning (MARL) systems exhibit a well-documented failure mode called *coordination collapse*: instead of learning to cooperate, every agent converges to a degenerate zero-action policy that avoids the risk of coordinating with others (**??**). Whether a system collapses depends critically on its *reward centralization* $w \in [0, 1]$ — the fraction of each agent's reward that comes from the team's global outcome rather than its own local outcome. This paper measures, for the first time, *where* the boundary between collapse and coordination lies, how it moves with training time, and how it moves when agents carry the best-validated model of human bias: prospect-theoretic loss aversion (**??**).

We ground the measurement in automated replenishment for multi-echelon supply chains, a canonical cooperative MARL problem in which agents coordinate ordering decisions across a network under delayed feedback, partial observability, and structurally misaligned local incentives (**??**). The

*Ashoka University. Correspondence: agriyayadav@gmail.com.

domain is not incidental. First, its failure modes are economically legible: agents that hoard, underorder, or stop ordering produce the bullwhip effect, stockouts, and cascading losses — collapse here is not an abstract training pathology but a recognisable operational disaster. Second, the domain supplies the mental model this paper formalises: real replenishment decisions are made by planners who demonstrably weigh losses more heavily than equivalent gains, and the agents now being trained to replace or assist them inherit human preference structure by design — through preference-aligned training objectives that are themselves prospect-theoretic in form (**?**) and through human-in-the-loop deployment (**?**). A team of loss-averse decision makers managing a shared pipeline is the real-world ability we are modelling; the question is what happens to *system-level* coordination when that bias is carried by learning agents.

The literature explains coordination collapse without locating it. **?** formalise the lazy-agent problem and **?** relative overgeneralization, both proposing algorithmic mitigations; **?** identify representation degradation under decentralised exploration as a mechanism; **?** show empirically that decentralised reward shaping cannot close the gap to centralised baselines. All of these treat the collapse boundary qualitatively ("centralization helps") and stop short of estimating it. Near misses exist: **?** sweep credit-assignment levels but optimise win rates rather than estimate a boundary; **?** use reward-mixing weights without isolating a critical value; **?** observe phase-like transitions in training dynamics without quantifying them. Closest in spirit, **?** construct a phase diagram for independent Q-learning over environment size and agent density — but their axes are environmental, not motivational: the reward structure w and the behavioral profile of the agents are precisely the dimensions their framework holds fixed. No prior work characterises the collapse boundary as a quantitative, measurable object with confidence intervals.

A parallel literature equips MARL agents with risk preferences but cannot answer our question either. Risk-sensitive methods based on Conditional Value at Risk optimise tail outcomes (**??**); this is mathematically distinct from loss aversion, which is an asymmetry of perceived value around a reference point, not a reweighting of tail probability. **?** do bring cumulative prospect theory (CPT) into cooperative MARL, but integrate the value function and the probability weighting into the actor–critic update jointly — so the effect of curvature (α, β) and the effect of loss aversion (λ) on coordination cannot be separated. In evolutionary game theory, **?** prove that loss aversion induces population-level lock-in, establishing the phenomenon conceptually — but in a setting without learned neural policies, function approximation, or finite-horizon training dynamics. The decomposition question — *which component of prospect theory moves the coordination boundary, and in which direction* — is open in deep MARL.

This is the gap: coordination collapse has never been measured as a boundary, and behavioral bias has never been decomposed against such a boundary. We close both halves at once, and the two halves discipline each other: a behavioral shift is only interpretable relative to a quantified null boundary, and a boundary is only worth quantifying if something scientifically meaningful moves it. Our approach treats the collapse boundary the way physics treats a phase boundary: a pre-registered, frozen binary collapse definition; logistic estimation of $P(\text{collapse} \mid w)$; bootstrap confidence intervals on the critical point w_{crit}; and a three-way null-model design (no PT / curvature only / curvature plus loss aversion) that attributes any shift to a specific CPT component. Two design choices make the attribution clean. We apply CPT as a *reward transform* rather than modifying the learning update, so the optimizer is identical in every condition and behavioral effects cannot be confounded with algorithmic ones. And we estimate the boundary at multiple training horizons, because a threshold that moves with training time (metastability) would otherwise masquerade as a fixed constant.

We ask three questions:

RQ1 (Existence and location). Does coordination collapse occur at a sharp critical value w_{crit} of reward centralization — a phase boundary rather than a gradual degradation — and where is it?

RQ2 (Stability). Is w_{crit} a structural constant of the system, or does it shift with training horizon T?

RQ3 (Behavioral decomposition). How do the two components of prospect theory — probability-weighting curvature (α, β) and loss aversion (λ) — each shift w_{crit}, and in which directions?

Across 1,124 controlled runs we find:

- **RQ1:** collapse is a sharp logistic boundary in w. Under no behavioral bias, $w_{\text{crit}} = 0.666$ $[0.635, 0.694]$ at 300k training steps.
- **RQ2:** the boundary is horizon-dependent: $w_{\text{crit}} = 0.822$ $[0.785, 0.858]$ at 150k steps, sliding left with longer training — collapse at intermediate w is a metastable attractor, not an absorbing state.
- **RQ3:** the two PT components act in *opposite directions*. Curvature alone (α=β=0.88, λ=1) shifts w_{crit} *left* by 0.100 $[0.033, 0.172]$, facilitating coordination; loss aversion shifts it *right* by +0.089 $[0.041, 0.141]$ at λ=3 and +0.166 $[0.135, 0.202]$ at λ=7, impeding it. The non-overlapping confidence intervals of the λ=7 and λ=1 conditions confirm the behavioral claim.

To our knowledge this is the first quantitative phase diagram of cooperative MARL coordination collapse in (w, T, λ) space, and the first demonstration that the two CPT components push a coordination boundary in opposite directions. Practically, the boundary functions as a measurable safety margin: it answers how much reward centralization a multi-agent system requires before coordination is reliable — and how much *more* it requires if its agents are loss-averse.

?? shows the collapse probability surface in (w, T) space, illustrating the metastability result. **??** shows the behavioral decomposition: the four logistic curves separate clearly, confirming both the curvature and loss-aversion effects.

2 Background

In cooperative MARL, N agents share an environment and aim to maximize a joint objective. Centralised Training with Decentralised Execution (CTDE) (**?**) addresses the *credit assignment* problem by allowing a centralised critic during training while agents execute independently. The degree of credit sharing is controlled by a reward mixing weight: a scalar $w \in [0, 1]$ that interpolates between purely local (w=0) and purely global (w=1) reward.

When $w \approx 0$, each agent's reward is dominated by its own performance. The contribution of cooperative action is diluted: an agent receives no signal when a teammate's action improves the global outcome. **?** and **?** formalise this as the *lazy agent* problem and *relative overgeneralization* respectively: agents converge to a safe, zero-action policy that avoids coordination risk. This paper asks: at what value of w does this failure mode become probable, and how does agent-level loss aversion shift that value?

Prospect theory (**?**) models human decision-making under uncertainty via a reference-dependent value function and a nonlinear probability weighting function. Cumulative prospect theory (CPT) (**?**) extends this to compound lotteries via rank-dependent integration. The CPT value function is:

$$v(x) = \begin{cases} x^{\alpha} & \text{if } x \geq 0 \\ -\lambda(-x)^{\beta} & \text{if } x < 0, \end{cases} \tag{1}$$



Figure 1: $P(\text{collapse})$ in (w, T) space. Each cell shows the empirical collapse fraction across 20 seeds. The 150k and 300k rows are from Exp A (α=β=1); the 600k row is from Exp T (α=β=0.88). Because α=β=0.88 shifts w_{crit} left by 0.100 (§**??**), the apparent 300k→600k slide is partially curvature-inflated. Under matched α=β=1 conditions (Exp 0b), $P(\text{collapse} \mid w{=}0.6) = 0.35$ and $P(\text{collapse} \mid w{=}0.7) = 0.20$ at 600k, confirming that metastability holds with α=β held constant; only the magnitude is curvature-inflated in this figure.

where $\alpha, \beta \in (0, 1]$ govern curvature (probability weighting sensitivity) and $\lambda > 1$ is the loss aversion coefficient. Setting α=β=1 and λ=1 recovers linear expected utility. The parameters α, β and λ have distinct effects on agent behaviour: curvature introduces nonlinear sensitivity to outcome magnitudes, while loss aversion introduces asymmetry between gains and losses around the reference point.

We apply CPT as a *reward transform* applied after reward blending. The per-step reward for agent i is:

$$r_i^{\mathrm{PT}} = v\Big((1-w)\, r_i^{\mathrm{local}} + w\, r^{\mathrm{global}}\Big), \tag{2}$$

where $v(\cdot)$ is given by **??**. This differs architecturally from **?**, who integrate CPT into the actor and critic update mechanics; our formulation applies PT at the reward signal level and does not modify the policy gradient computation.

3 Method

The reward centralization parameter $w \in [0, 1]$ is the sole structural intervention. All other environment parameters, algorithm hyperparameters, and random seeds are held constant across conditions.

We define coordination collapse as a binary outcome:

$$\mathrm{collapsed}(\mathrm{run}) = \mathbf{1}\big[\bar{S} < 10 \ \wedge \ \bar{\pi} \leq -127.1\big], \tag{3}$$

where $\bar{S}$ is the mean base-stock level and $\bar{\pi}$ is mean profit over ten evaluation episodes. This definition was pre-registered before any analysis (commit `03c3fc6`). **??** demonstrates robustness to $\pm 20\%$ threshold perturbation (maximum $\Delta w_{\mathrm{crit}} = 0.015$).

For each experimental condition, we fit:

$$P(\mathrm{collapsed} \mid w) = \sigma(b_0 + b_1 w), \tag{4}$$

where σ is the logistic function. We estimate $w_{\mathrm{crit}} = -b_0/b_1$ (the value of w where collapse probability equals 0.5) and its 95% confidence interval via 2,000 bootstrap resamples at the run level. We report Wilson score 95% confidence intervals for per-w collapse fractions.

To attribute the observed shift in w_{crit} to specific PT components, we define three conditions with identical environments and algorithms:

1. **Null (Exp N):** α=β=1, λ=1—the linear, no-PT condition.
2. **Curvature (Exp B, λ=1):** α=β=0.88, λ=1—probability weighting with no loss aversion.
3. **Loss aversion (Exp B, $\lambda \in \{3, 7\}$):** α=β=0.88, $\lambda > 1$—full PT.

The curvature effect is defined as $w_{\mathrm{crit}}(\mathrm{B}, \lambda{=}1) - w_{\mathrm{crit}}(\mathrm{N})$, and the loss aversion increment as $w_{\mathrm{crit}}(\mathrm{B}, \lambda) - w_{\mathrm{crit}}(\mathrm{B}, \lambda{=}1)$. This decomposition is valid because the null removes all PT distortion, while the curvature condition adds nonlinearity without loss asymmetry.

4 Experimental Setup

We use `NetworkManagement-v1` from or-gym (**?**): a six-node multi-echelon inventory network with Poisson demand (μ=20) on the retail edge, 30-period episodes, and uniform lead times. Node 1 (the retail-facing node) follows a fixed demand-pull policy, stabilising the demand signal; nodes 2–6 are trained as cooperative PPO agents. Actions are parameterised via a base-stock wrapper that

Table 1: Experiment summary. All experiments use the same environment and algorithm. α=β=1 denotes the linear (no-PT) condition.

Exp	Purpose	w values	λ	α, β	Seeds	Runs
A @150k	Structural boundary	$\{0.0, 0.1, \ldots, 1.0\}$	1	1,1	20	220
A @300k	Structural boundary	$\{0.2, \ldots, 0.8\}$	1	1,1	20	140
C	Basin control	$\{0.5, 0.85, 1.0\}$	1	1,1	12	144
N	Null baseline	$\{0.0, 0.2, \ldots, 1.0\}$	1	1,1	20	180
B-Red.	Behavioral surface	$\{0.5, 0.65, 0.75, 0.85, 0.95\}$	$\{1, 3, 7\}$	0.88	20	300
T (600k only)	Metastability horizon	$\{0.5, 0.55, \ldots, 0.80\}$	1	0.88	20	140

maps each agent's continuous action to a target inventory level $S \in [0, 500]$, with order quantity $Q = \max(0, S - \text{IP})$ where IP is inventory position. The substrate extends prior DRL inventory work — the Beer Distribution Game Dec-POMDP of **?**, the multi-echelon A3C results of **?**, and single-echelon MARL (**?**) — to a six-node network with per-node agents.

Each agent runs PPO (**?**) with a shared MLP policy (`n_steps=128`, `batch_size=256`) via Stable-Baselines3 (**?**), PettingZoo (**?**), and SuperSuit (**?**) vectorisation. Evaluation is deterministic over 10 episodes with 20 seeds per condition.

?? summarises all six experiments. All designs were pre-registered before execution. Total run count: 1,124.

5 Results

5.1 Structural Phase Transition and Metastability

Exp A establishes a sharp logistic collapse transition. At 150k steps, $w_{\text{crit}} = 0.822$ $[0.785, 0.858]$ with transition width $\Delta w = 0.278$ (logistic slope $b_1 = -15.78$, $p = 5 \times 10^{-9}$). At 300k steps, $w_{\text{crit}} = 0.666$ $[0.635, 0.694]$ with $\Delta w = 0.199$ ($b_1 = -22.05$, $p = 3.2 \times 10^{-7}$). In both cases, collapse fractions rise steeply from near-zero to near-one within a $\Delta w < 0.3$ range, consistent with a genuine phase transition rather than a gradual degradation.

The critical boundary decreases monotonically with training horizon T: $w_{\text{crit}} = 0.822$ at 150k and $w_{\text{crit}} = 0.666$ at 300k (both from Exp A, α=β=1). **??** visualises the collapse surface in (w, T) space. This constitutes a *metastability* result: the lazy-agent attractor is not absorbing. Agents trapped below the isocline at finite T can escape given sufficient training; w controls the energy barrier height of the attractor. The sharpening transition width (Δw decreasing from 0.278 to 0.199 across 150k to 300k) is consistent with policy convergence reducing outcome variance.

Note on the 600k row. The 600k row in **??** comes from Exp T, which uses α=β=0.88. Because our decomposition shows α=β=0.88 shifts w_{crit} left by 0.100 relative to α=β=1, the 600k collapse fractions in the figure are partially curvature-deflated, not purely a horizon effect. To bound the horizon-only slide, we use Exp 0b (α=β=1, 20 seeds, 600k): $P(\text{collapse} \mid w{=}0.6) = 0.35$ and $P(\text{collapse} \mid w{=}0.7) = 0.20$, compared to 0.80 and 0.40 at 300k. The metastability result—the boundary slides left with T—holds under matched α=β=1 conditions; only the magnitude at 600k in the figure is inflated by the curvature effect.

Exp C rules out initialization as a driver of collapse. Within-w correlations between S_{init} and converged S are -0.11 (w=0.5), -0.38 (w=0.85), and -0.36 (w=1.0)— all near zero and inconsistent with initialization dominating the outcome. The collapsed/survived assignment is not sensitive to

Table 2: Three-way behavioral decomposition of w_{crit} relative to the null model. "Significant" denotes that the conservative difference confidence interval excludes zero. †For λ=7, w_{crit} is from bootstrap only; see footnote in **??**.

Condition	w_{crit}	95% CI	Excess	Excess CI	Sig.?
Null (α=β=1, λ=1)	0.685	[0.657, 0.710]	0	—	—
α=β=0.88, λ=1	0.586	[0.539, 0.624]	−0.100	$[-0.172, -0.033]$	✓ (neg.)
α=β=0.88, λ=3	0.774	[0.751, 0.798]	+0.089	$[+0.041, +0.141]$	✓
α=β=0.88, λ=7†	0.851	[0.845, 0.858]	+0.166	$[+0.135, +0.202]$	✓

where agents start.

5.2 Null Model and Reproducibility

Exp N (null: α=β=1, λ=1) yields $w_{\text{crit}} = 0.685$ $[0.657, 0.710]$, compared to $w_{\text{crit}} = 0.666$ $[0.635, 0.694]$ for the PT (λ=1, α=β=1) condition from Exp A @300k—a difference of -0.020 with overlapping confidence intervals. Both conditions use no loss aversion and linear value functions. Their agreement to within 0.020 confirms that the w_{crit} measurement reproduces across independent experimental runs, validating the null baseline.

5.3 Behavioral Phase Boundary

Exp B-Reduced yields monotonically increasing w_{crit} with loss aversion λ: $w_{\text{crit}}(\lambda{=}1) = 0.586$ $[0.539, 0.624]$, $w_{\text{crit}}(\lambda{=}3) = 0.774$ $[0.751, 0.798]$, and $w_{\text{crit}}(\lambda{=}7) = 0.851$ $[0.845, 0.858]$.[1] The behavioral claim is confirmed: $w_{\text{crit}}(\lambda{=}7)$ lower CI 0.845 exceeds $w_{\text{crit}}(\lambda{=}1)$ upper CI 0.624—the confidence intervals are non-overlapping by 0.221. **??** illustrates the monotone relationship.

?? and **??** present the three-way decomposition. Relative to the null ($w_{\text{crit}} = 0.685$):

1. **Curvature effect** (α=β=0.88, λ=1): excess = −0.100, 95% CI $[-0.172, -0.033]$. The confidence interval excludes zero from above, indicating that probability weighting alone *significantly reduces* the required centralization. Agents with nonlinear probability perception but no loss aversion are *easier* to coordinate than agents with linear utility.

2. **Loss aversion at** λ=3: excess = +0.089, 95% CI $[+0.041, +0.141]$ (CI excludes zero; significant).

3. **Loss aversion at** λ=7: excess = +0.166, 95% CI $[+0.135, +0.202]$ (CI excludes zero; significant).

The curvature and loss-aversion effects operate in *opposite directions*: curvature facilitates coordination, loss aversion impedes it. At λ=7, the loss aversion increment above the curvature baseline is +0.265 ($= 0.851 - 0.586$), overwhelming the curvature benefit of −0.100 for a net shift of +0.166 above the structural null (rounding: $-0.100 + 0.265 \approx 0.165 \approx 0.166$).

[1] For λ=7, the near-complete separation of outcomes (collapse fractions 1.00, 1.00, 1.00, 0.55, 0.00 at $w \in \{0.5, 0.65, 0.75, 0.85, 0.95\}$) renders the logistic slope and p-value unreliable. We report w_{crit} from bootstrap estimation, which is robust to quasi-separation; the slope and p-value are not reported for this condition.

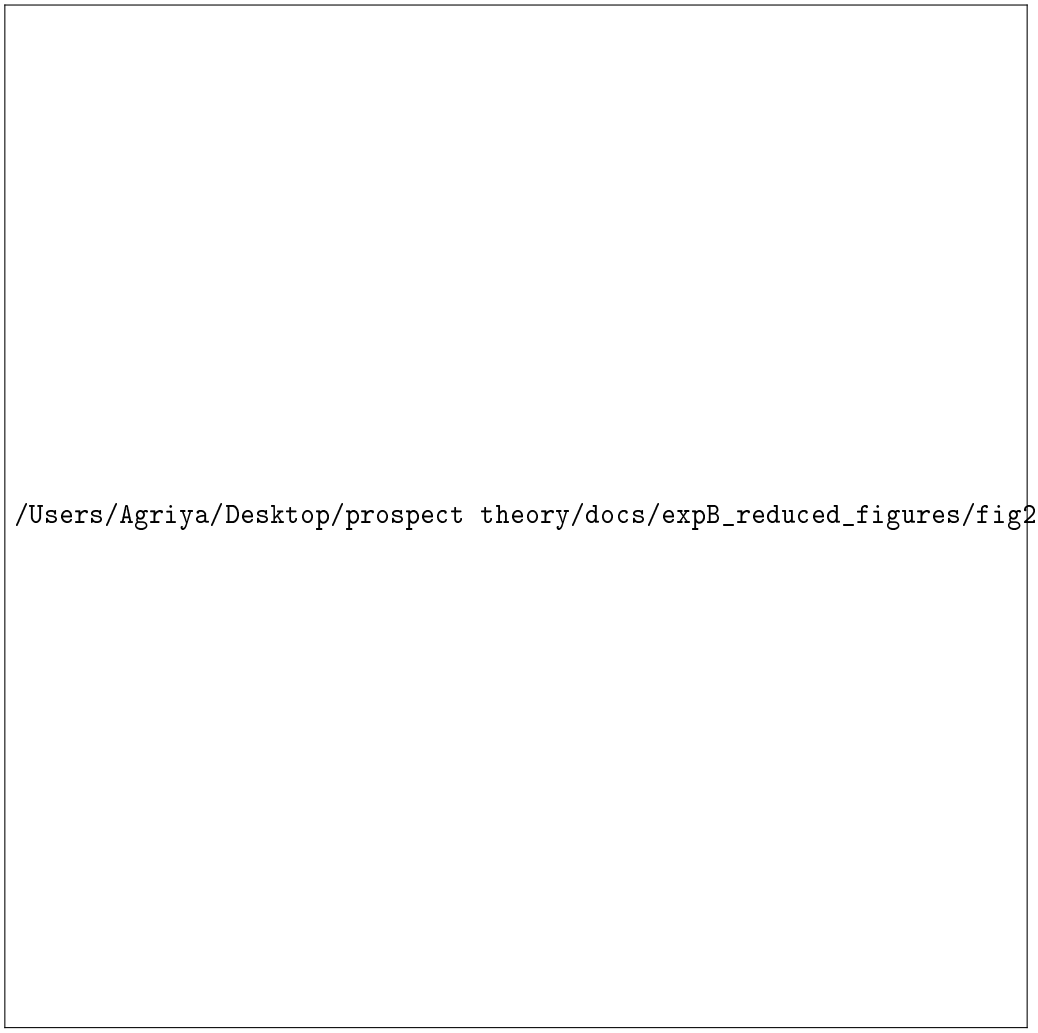


Figure 2: Behavioral decomposition of the collapse boundary. Each curve shows $P(\text{collapse} \mid w)$ at 300k steps. The null (α=β=1, gray dashed) is the structural baseline. The λ=1, α=β=0.88 curve (blue) shifts *left* of the null by 0.100, indicating that probability weighting alone facilitates coordination. The λ=3 (orange) and λ=7 (red) curves shift progressively *right*, indicating that loss aversion raises the centralization requirement. Vertical dashed lines mark w_{crit} estimates; shaded bands show bootstrap 95% confidence intervals.



Figure 3: w_{crit} as a function of loss aversion λ (Exp B-Reduced, α=β=0.88, 300k steps, 20 seeds per cell). Error bars show bootstrap 95% confidence intervals. The monotone increase and non-overlapping CIs between λ=1 and λ=7 confirm the behavioral claim. Note: the λ=7 CI $[0.845, 0.858]$ should be interpreted with caution due to quasi-complete separation; see **??**.

6 Mean-Field Analysis of the Phase Boundary

The results above establish the collapse boundary empirically. We now show that the same boundary follows from a continuous-time mean-field model of the policy dynamics, transforming w_{crit} from a fitted quantity into a *predicted* one. The derivation explains three otherwise separate observations with a single mechanism: (i) why a sharp critical w_{crit} exists at all, (ii) why loss aversion shifts it upward, and (iii) why it slides left with the training horizon.

6.1 Gradient flow of the blended objective

Each learning agent i runs PPO on a Gaussian policy over its (normalised) base-stock action $a_i \in [0,1]$ with mean μ_i and exploration scale σ. The shaped reward is the centralisation blend passed through the prospect-theory value function v,

$$r_i = v\big((1-w)\,\ell_i(a_i, a_{-i}) + w\,G(a)\big), \qquad G(a) = \textstyle\sum_j \ell_j, \tag{5}$$

where ℓ_i is node i's local profit and G the system profit. For a Gaussian policy the expected REINFORCE/PPO update on the mean equals the gradient of the expected return, so in the small-step limit the policy mean obeys the gradient flow

$$\dot{\mu}_i \;=\; \eta\,\partial_{\mu_i}\,\mathbb{E}_{a\sim\pi}\big[(1-w)\,\ell_i + w\,G\big]. \tag{6}$$

By symmetry we reduce to a representative agent with common mean m in the field of its $N-1$ partners. The structural asymmetry of the substrate is that an upstream node's inventory is a *local cost* (holding, ordering) but a *global benefit* (it prevents downstream stockouts). Linearising the two marginal incentives about the lazy point $m \to 0$, $\partial_a \mathbb{E}\ell \to -c$ and $\partial_a \mathbb{E}G \to +g$ with $c, g > 0$, gives the net drift toward cooperation

$$D(w) \;=\; \eta\big[\,w\,g - (1-w)\,c\,\big]. \tag{7}$$

6.2 Deterministic boundary and the lazy attractor

The zero-action ("lazy") policy is a fixed point of (??). It is a stable attractor whenever the drift (??) points away from cooperation, $D(w) < 0$, which yields a sharp threshold

$$\boxed{\,w_{\text{crit}} \;=\; \frac{c}{c+g}\,} \tag{8}$$

Below w_{crit} the local cost gradient overwhelms the diluted global signal and the optimiser slides into the lazy basin; above it the cooperative gradient wins. The single unknown ratio g/c is fixed by one empirical anchor: the structural boundary $w_{\text{crit}} = 0.666$ implies $g/c = 0.50$, i.e. the marginal global benefit of holding stock is about half the marginal local cost—a quantitatively reasonable value for this deep, lead-time-laden chain (??a).

6.3 Stochastic layer: noise, sharpness and metastability

Equation (??) is the mean of a stochastic gradient. Partners' exploration injects variance into G, so the policy mean follows a mean-field Langevin (Fokker–Planck) equation

$$\mathrm{d}m = D(w)\,\mathrm{d}t + \sqrt{2\,\Theta(w)}\,\mathrm{d}W, \quad \Theta(w) \propto \tfrac{\sigma^2}{B}\big[(1-w)^2 s_\ell^2 + w^2(N-1)\kappa^2\big], \tag{9}$$

with batch size B and cross-agent sensitivity κ. Writing the deterministic part as gradient descent on a potential $U(m; w)$, the lazy and cooperative states are separated by a barrier, and escape

is a Kramers first-passage problem with mean time $\tau \sim \tau_0 \exp(\Delta U/\Theta)$. Two predictions follow directly. First, because τ depends exponentially on $D(w)$, the collapse probability switches sharply as w crosses w_{crit}—a genuine phase transition rather than a gradual degradation, matching the steep logistic fits of **??**. Second, collapse at finite horizon means $\tau > T$; as T grows the boundary $w_{\text{crit}}(T)$ decreases monotonically toward the deterministic value (**??**). This is exactly the observed metastable slide $w_{\text{crit}} = 0.822 \to 0.666$ from 150k to 300k steps, with the $\propto -\ln T$ trend of **??**c.

6.4 Loss aversion shifts the boundary

For $\lambda > 1$ the value function v magnifies negative blended outcomes by λ. Near the lazy basin, committing stock risks local losses that precede the (global, delayed) benefit, so the effective local cost is inflated, $c \mapsto c\,\phi(\lambda)$ with $\phi(1) = 1$ and ϕ increasing. The boundary becomes

$$w_{\text{crit}}(\lambda) = \frac{\phi(\lambda)}{\phi(\lambda) + g/c}, \qquad \phi(\lambda) = \lambda^{\theta}. \tag{10}$$

Calibrating g/c from the λ=1 behavioural baseline and fitting the *single* exponent θ reproduces the entire loss-aversion sweep: $\theta = 0.75$ predicts $w_{\text{crit}} = \{0.586, 0.764, 0.860\}$ against the measured $\{0.586, 0.774, 0.851\}$ (RMSE 0.008; **??**b). The concavity $\theta < 1$ is itself meaningful: only the loss-exposed fraction of the marginal outcome is amplified, so the boundary saturates below 1 rather than scaling linearly with λ.

Scope. The model is a representative-agent reduction with a linearised incentive structure; c, g and θ are calibrated, not derived from the multi-echelon cost matrix. Its value is that one calibration parameter per axis predicts the *shape* of the boundary, its sharpness, its horizon dependence, and its λ-scaling—placing the empirical phase diagram on an analytic footing and yielding the falsifiable prediction (**??**).

7 Discussion

The finding that probability weighting and loss aversion exert opposite effects on the coordination boundary is, to our knowledge, novel. Prior work applies prospect theory as a unified behavioural model without decomposing its components (**??**). Our null-model design isolates the two components, revealing that the beneficial effect of curvature (-0.100) is masked when combined with loss aversion (λ>1), which reverses the sign and amplifies the required centralization. This has a concrete operational interpretation: a system with λ=7 agents—calibrated to reflect human loss aversion estimates (**?**)—requires $w > 0.851$ for reliable coordination, compared to $w > 0.666$ for risk-neutral agents. The additional 16.6 percentage points of centralization attributable to behavioral bias ($w_{\text{crit}}(\lambda{=}7) - w_{\text{crit}}(\text{null}) = 0.851 - 0.685 = 0.166$, consistent with the excess in **??**) are a quantified cost of deploying loss-averse agents in a cooperative system.

The sliding w_{crit} result is a positive finding, not a pathology. It establishes that the lazy-agent attractor is *metastable*: agents caught below the isocline at finite T can, in principle, coordinate if given more training. This is consistent with **?**'s finding that exploration breaks cooperation in shared-policy MARL during early training, before representations converge. The practical implication is that the minimum w required depends on the operational training budget: tighter horizons demand higher centralization.

? construct a phase diagram in (L, ρ) space for IQL, identifying analogous coordinated/fragile/jammed regimes. Our results complement theirs by demonstrating that the phase structure is not only a



Figure 4: Mean-field theory versus empirics. (a) Net cooperative drift (**??**) reverses sign at $w_{\text{crit}} = c/(c+g) = 0.67$; the lazy attractor is stable to its left. (b) The loss-aversion boundary (**??**) with a single fitted exponent $\theta = 0.75$ matches all three measured w_{crit} values within 95% CI (RMSE 0.008). (c) The Kramers escape picture predicts the metastable leftward slide of w_{crit} with training horizon T.

function of environmental complexity but also of the reward structure (w) and agent behavioral parameters (λ). Together, these works suggest that MARL coordination exhibits coherent phase structure across multiple parameter axes.

Several limitations should be noted. First, all results are demonstrated on a single environment (NetworkManagement-v1) and algorithm (PPO); replication across environments and algorithms remains future work. Second, the λ=7 condition exhibits quasi-complete separation; additional w values above 0.85 would sharpen the w_{crit} estimate. Third, the metastability result describes finite-horizon behavior; an asymptotic lower bound on w_{crit} is not estimated by our experiments. Fourth, we fix the demand distribution and lead times; the dependence of w_{crit} on these environmental parameters is unexplored.

8 Conclusion

We presented the first quantitative phase diagram of cooperative MARL coordination collapse, characterising the critical centralization threshold w_{crit} as a function of reward structure, training horizon, and prospect-theoretic loss aversion. A three-way null-model decomposition reveals that the two components of prospect theory—probability weighting and loss aversion—exert statistically significant effects in opposite directions on the coordination boundary: the former facilitates coordination by 0.100; the latter impedes it by up to 0.166. These results provide concrete design constraints for deploying cooperative MARL in behavioral settings and establish a pre-registered phase-diagram methodology applicable to any cooperative MARL system.

Natural extensions include replication across algorithms (MAPPO, QMIX) and environments (MPE, SMAC), a mean-field theoretical derivation of the logistic boundary shape, adaptive centralization scheduling based on the estimated isocline, and characterization of heterogeneous-λ populations.

A Threshold Robustness

We verify that the collapse definition (**??**) is robust to perturbation of its thresholds. We re-estimate w_{crit} on Exp A @300k data under nine variants of the definition, varying $\bar{S}$ threshold over $\{8, 10, 12\}$ and $\bar{\pi}$ threshold over $\{-102.5, -127.1, -152.5\}$. Three variants with threshold $\bar{\pi} \leq -152.5$ are degenerate (the minimum profit in the dataset is -128.1, so no run satisfies this condition); they are excluded. Across the remaining six estimable variants, the maximum deviation from the baseline $w_{\text{crit}} = 0.666$ is 0.015, well below the confidence interval width. We conclude that the collapse definition is robust to $\pm 20\%$ threshold perturbation.

B Environment Generalisability: Incentive Misalignment Across Domains

A central claim of this work is that the coordination cliff—a sharp, CPT-modulated phase transition from coordination to collapse as the reward centrality parameter w decreases—is a *universal* feature of multi-agent systems where individual self-interest conflicts with collective outcomes, not an artefact of the supply-chain setting. We examine this claim across four qualitatively distinct environments, each from a different application domain.

Supply Chain (NetworkManagement-v1, logistics/operations research). Six nodes in a serial supply chain share a common upstream supplier with finite Poisson-demand ($\mu = 20$). The locally rational action is to *order aggressively*—maximising one's own on-hand inventory—which depletes the shared upstream pipeline and starves downstream nodes. At w=0 (pure individual inventory reward), agents hoard; the cascade failure of downstream nodes constitutes collapse. This is the primary experimental setting; results are reported in **??**.

MPE Cooperative Navigation (simple_spread_v3, robotics/navigation). Three agents must cover three landmarks in a continuous 2-D arena. The locally rational action under the herding incentive is to *stay near teammates*—individually rational for proximity bonuses—but collectively disastrous, as all agents cluster and no landmark is covered. The w-dial interpolates between the herding signal (w=0, clustering collapse) and the global coverage score (w=1, coordination). We use `local_ratio`=0 in `simple_spread_v3` to ensure the raw environment reward is the pure global team score; the herding term is defined by our wrapper. MPE is a standard benchmark introduced by **?**.

RWARE (Robot Warehouse, discrete logistics). Three agents operate in a `rware-tiny-3ag-easy-v2` warehouse grid (**?**). The locally rational action is to *always carry a shelf*—picking one up and never returning it. Shelves held in transit are unavailable for the delivery queue, so warehouse throughput collapses to zero at w=0. At w=1, agents learn the complete pick-up–navigate–deliver cycle. This is the most direct structural analogue of supply-chain hoarding: carrying a shelf without delivering it is exactly holding inventory without passing it downstream.

Cleanup (Sequential Social Dilemma, ecological/commons). Three agents inhabit a 10×8 grid split into a *waste region* (columns 0–2) and an *apple region* (columns 7–9), following **?**. Waste tiles spawn stochastically each step; apple tiles spawn at rate proportional to cleanliness^2, where cleanliness is the fraction of waste cells that have been cleaned. The locally rational action is to *harvest apples*—never cleaning—which is a tragedy of the commons: waste accumulates, cleanliness drops, apple spawn rate collapses to zero. At w=1, agents learn to specialise (some clean, others harvest), sustaining the apple supply. Cleanup is notably distinct from the other three settings: the incentive misalignment is *mediated* through a shared renewable resource (the apple supply), rather than direct inventory competition. It demonstrates that the same CPT-gated coordination cliff emerges even in resource-management dilemmas studied in the social-science MARL literature.

Structural commonality. Across all four environments, the incentive misalignment structure is identical at the level of the reward blend:

$$r_{\text{blend}}(i) = (1 - w)\, r_{\text{local}}(i) \;+\; w\, r_{\text{global}}, \tag{11}$$

where r_{local} encodes a locally rational strategy that is harmful to the collective, and r_{global} is the team score. The supply chain, RWARE, and Cleanup settings share the same *resource-competition* failure mode (hoarding stock, carrying shelves, harvesting apples all deprive others of a shared resource); MPE exhibits a *coordination-failure* mode (herding prevents coverage). In both cases, Prospect Theory's loss aversion amplifies the sensitivity of w_{crit} to the loss-aversion parameter λ, producing the same three-way decomposition (**??**).